

Unit 4-Review

Motion

1. The position of an object moving along a straight line is described by the function $s(t) = -t^3 + 4t^2 - 10$ for $t \geq 0$.
Is the object speeding up or slowing down when $t = 3$?
2. A position function of an object is given by: $s(t) = t^3 - 6t^2 + 8t$, $t \geq 0$
 - (a) Determine the velocity function for the object.
 - (b) Identify the point(s) where the object is at rest.
 - (c) Identify the point(s) where the acceleration is zero.
 - (d) Determine the equation of the acceleration function.
 - (e) For which intervals is the acceleration negative? Positive?
 - (f) Determine the intervals for which the object is speeding up and slowing down.

Optimization

Perimeter and Area

1. A farmer has 2400 m of fencing to create a rectangular field that borders on a straight river. He does not need to fence the river edge. What will be the area of the largest possible field?
2. A man is building a fence around a portion of his backyard. He wants the area inside the fence to be 600 m^2 . If his house accounts for 20 m of the fence, what is the minimum amount of fencing he needs to purchase?

Volume and surface area

1. A box with a square base and open top has a volume of 4000 cm^3 . Determine the dimensions of the box that minimizes the amount of material required.
2. A cylindrical can is to be made to hold 1 litre of oil. Prove that a ratio of height to radius of 2:1 will minimize the cost of the metal to manufacture the can if the entire can is made of the same metal.
3. Find the circular cylinder of greatest volume that can be inscribed in a sphere of radius 6 cm.

Distances

1. Determine the point on the graph of $4x + y^2 = 0$ that is closest to the point $(0, -3)$.
2. A storage shed that is 3 m high and 2 m wide is built against a brick wall. Determine the length of the shortest ladder that will reach the wall from the ground, if it must go over the shed.

Best Time

1. An island is at a point A, 6 miles off shore from the nearest point, B, on a straight beach. A store is located at point C, 7 miles down the beach from B. If a man can row at the rate of 4 miles per hour and walk at the rate of 5 miles per hour, where should he land in order to get from the island to the store in the least possible time?
2. A canoeist is 300m offshore and wishes to land and then walk to a distant point on the straight shoreline. If she can paddle at 3 km/h and walk at 5 km/h, where should she land to minimize her travel time?

Profits, Cost, Revenue

1. Given a company's annual demand function is given by $p(x) = \frac{800000 - x}{200000}$ and cost function is $C(x) = 125000 + 0.42x$, where x is the number of widgets it produces. What level of sales will maximize profits?
2. A steel storage tank for propane gas is to be constructed in the shape of a cylinder with a hemisphere on each end. If the desired capacity is 100 m^3 , what dimensions will require the least amount of steel? **Note:** The cylinder requires steel material on its ends.
3. A metal can is to hold 1 L of oil. Determine the radius of the can that will minimize the cost of the metal.

Additional Questions

1. Determine the dimensions of the rectangle of maximum area that can be inscribed above the x -axis and below the parabola $y = 4x - x^2$.
2. Consider the curve $y = 10 - x^2$ for $x \geq 0, y \geq 0$. A tangent to this curve will form a triangle with the x -axis and the y -axis. Determine the equation of the tangent that forms a triangle of minimum area.